Demo: What the Coil Can Do

The payoff demonstrations — adult-operated

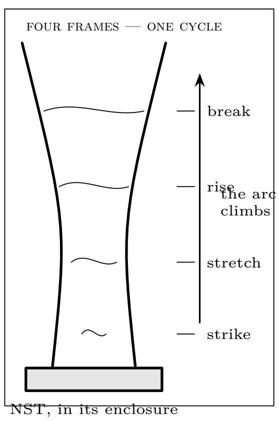
Jacob's ladder, and lighting a tube from your hand

Lesson Demo E · after the coil is built · **adult owns the mains side**

ADULT-OPERATED: MAINS-DERIVED HIGH VOLTAGE

From here on the apparatus contains voltages and currents that kill people every year. The adult owns the mains plug, the transformer and every decision about energizing. Nobody touches anything inside the enclosure until the supply is unplugged *and* the capacitor has been shorted with the grounding stick. Read the **Safety** sheet completely before this lesson, not during it.

Jacob's ladder

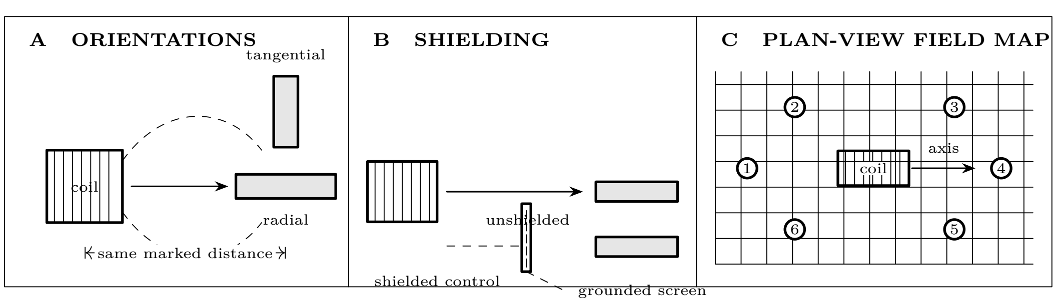


Predict first

Two wires make a narrow V, closest together at the bottom. An arc strikes across the gap and then travels *upward*, into a wider and wider gap that ought to be harder to jump. Why does it climb rather than staying where it started?

Why it strikes low	The gap is narrowest at the bottom, so that is where the field is strongest and the air breaks down first. *see the **10. The Spark Gap** sheet*
Why it climbs	The arc is a column of hot ionised gas, and hot gas rises. It also carries current, so it is pushed by its own magnetic field. Both effects drive it up the electrodes.
Why it survives	Ionised air conducts far better than cold air, so once an arc exists it can be stretched much wider than the gap it could originally jump.
Why it breaks	Eventually the arc is stretched too far, cools, and stops conducting. The voltage then reappears across the narrow bottom gap and the cycle restarts.
Build note	Two stiff bare wires, 1–2 mm apart at the base, opening to 5–8 cm at the top, mounted on a fireproof insulating base inside the enclosure.

Lighting a fluorescent tube from your hand



What happens	Stand several feet from a running coil holding an ordinary fluorescent tube, and it lights — with no wires and nothing connected to it.
Why	The coil floods the room with a strong, rapidly changing electric field. That field drives the free electrons in the tube's low-pressure mercury vapour back and forth, they collide with mercury atoms, the atoms emit ultraviolet, and the phosphor coating converts that to visible light. Exactly the same process as in the ceiling fixture, with the field arriving through the air instead of through a ballast.
It is a field meter	Walk toward the coil and it brightens; step behind a grounded metal sheet and it dims. You are mapping the field you drew with iron filings in Lesson 2, at room scale.
Distance rule	Do this at a stated distance, agreed before the coil is switched on, and never with any part of you inside the streamer range.
Not a party trick	Everyone in the room needs to be beyond the longest streamer you have seen from this coil, plus a wide margin. The tube does not have to be close to work.

Run it as an investigation

FREEZE THE GEOMETRY BEFORE POWER IS APPLIED

The adult places the tube at taped floor marks, chooses its orientation, and returns behind the operating line. Nobody walks toward a live coil. Switch off, discharge, verify, reposition, and only then begin the next short run.

Questions to answer before each run

1. At one distance, will a radial or tangential tube glow more evenly? Draw the field direction you expect through the tube before choosing.
2. If the glow is caused by the field, what should a grounded screen change? What result would count against that explanation?
3. At equal-radius stations around the coil, should brightness depend only on distance? Which paired stations test the coil's symmetry?
4. For the ladder, which visible event marks breakdown, which marks hot-gas motion, and which marks loss of conduction?

Troubleshoot before interpreting

Observation	Power-off check
Tube never glows	Verify the tube in a normal fixture; confirm marked distance and orientation; do not move closer during a run.
Brightness will not repeat	Fix observer position and adaptation time; use the same tube; check the coil run duration and video for unequal operation.
Screen has no effect	Check that it is the same grounded screen in the same place and that the tube is actually behind its projected area.
Opposite stations disagree	Look for metal furniture, wiring, asymmetric topload geometry, and unequal station radii. Record the asymmetry.
Ladder strikes above the base	With power isolated and discharged, inspect the base gap, contamination, electrode alignment, and unintended sharp points.
Arc leaves the intended electrodes	Stop. The adult inspects enclosure, clearance, grounding, and insulation before any further run.

Synthesis proof

Claim — evidence — mechanism — limit

Claim: both showpieces are consequences of fields and matter, not exceptions to the course. **Evidence:** the ladder repeats a strike–rise–break cycle, while tube brightness changes reproducibly with position, orientation, and shielding. **Mechanism:** breakdown makes a conducting hot-gas path; buoyancy and electromagnetic forces move it until cooling ends conduction. Around the coil, a changing electric field drives charges in the low-pressure tube; geometry changes how strongly that field acts, and a grounded conductor redirects it. **Limit:** brightness scores are comparative, the tube is nonlinear, and the two forces on the ladder arc have not been separated.

Proof check. A defensible conclusion names the fixed source setting, run time, tube, observer, station radius, and orientation; shows repeats; preserves an anomalous trial; and connects each plotted change to one geometrical change. “It lit” is an observation. The controlled contrasts are the proof.

THESE TWO ARE THE REWARD, NOT THE POINT

Both demonstrations use the mains-derived high-voltage supply, and the full **Safety** sheet applies without exception: named adult on the plug, grounding stick in reach, capacitor shorted before any hand goes near anything, second person present, everyone with an implant out of the room, all electronics elsewhere, eye and ear protection, extinguisher in reach. Run in short bursts. Stop at the first sign of anything unexpected — a smell of hot varnish, an arc where there should not be one, a transformer getting warm.

Close the loop

Ask, before you switch anything on, which lesson each of these comes from. The ladder is Lesson 10 — air breaking down and becoming a conductor. The tube is Lesson 2 — a field reaching across empty space to do work. If they can name those without help, the course did its job, and the sparks are just the receipt.

Worksheets — complete after the runs

Ladder cycle record

Time from video recorded outside the operating boundary; do not approach the apparatus to read a scale.

Run	Strike height	Break height	Climb time	Cycles / 10 s	Arc path / sound
1					
2					
3					

Intermediate check. If one climb takes twice as long, inspect the video: did it strike at the same height and follow the same electrode? Repeat rather than silently averaging a different kind of cycle.

Tube field matrix

Use an agreed visual scale: 0 dark, 1 barely visible, 2 dim, 3 clear, 4 bright. The same observer scores every run from the same place.

Station	Distance	Orientation	No screen	Screen	Glow pattern / flicker
1		radial			
1		tangential			
2		radial			
3		radial			
4		radial			
5		radial			
6		radial			

Graph it twice. First graph mean brightness against distance, using matched orientation only. Then graph station number around one equal-radius ring. Put screened and unscreened results on the same axes. Label subjective brightness as an ordinal score, not a physical unit.